PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Morakinyo Bolatito
Program/Batch	B084D-Networkwalks
Date	12 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. networkwalks.com (lab-designated target in the Networkwalks program) 2. My own local lab network
Permission secured from client?	Yes (target designated in the Networkwalks lab instructions)
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local lab network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my Cybersecurity & Ethical Hacking program at Networkwalks.
The footprinting commands were run in Kali Linux, and Zenmap was used on my Windows PC for the scanning phase. Every step below includes the command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
ipconfig	Local IP address, subnet and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target. All commands were run on 11 November 2026.

First, I used **WHOIS** to obtain public domain registration information. The domain is registered through **GoDaddy** (created 2019-11-06, expires 2027-11-06), the registrant is hidden behind a privacy proxy, DNSSEC is unsigned, and the name servers are **NS6135.HOSTGATOR.COM** and **NS6136.HOSTGATOR.COM**, which identifies HostGator as the hosting provider.

I then used **WhatWeb** to identify the technologies used by the website. The results identified **Apache**, **WordPress 7.1.4** and **WP Download Manager 3.3.76**, along with jQuery 3.7.1, Bootstrap and Google Tag Manager. The HTTP site returned a 301 redirect to HTTPS, and a public contact email address (info@networkwalks.com) was exposed.

Using **Nslookup**, I resolved the domain name to its IP address, **192.232.216.135**. The resolver also printed a communication timeout after the answer, which was a local DNS issue and not a finding about the target.

I used **Curl** with the -I option to inspect the HTTP response headers. The request returned **301 Moved Permanently** to HTTPS, redirected by the WordPress "Really Simple Security" plugin, and showed **Server: Apache**, WordPress caching headers (X-nginx-cache, X-Endurance-Cache-Level) and an HttpOnly cookie. Permissions-Policy and Referrer-Policy headers were present.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)** after 2 requests.

Finally, I used **DNSRecon** to enumerate DNS records. The results showed the HostGator SOA and name servers, the mail server **mail.networkwalks.com**, an SPF record ending in **~all** (soft fail), a Google site verification TXT record, 8 cPanel autodiscover SRV records, and a DNS software version string (**BIND 9.16.23-RH**). The first run hit resolver timeouts and was repeated successfully.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local lab network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

On my Windows PC I entered the lab subnet **192.168.56.1/24** as the target in Zenmap and selected the **Ping Scan** profile (nmap -sn) to identify active hosts.

The scan, run on 12 November 2026, scanned **256 IP addresses** in 11.97 seconds and reported **1 host up**:

- 192.168.56.1

The Host Details tab shows the state as **up**, with no open, filtered or closed ports (a ping scan does not scan ports) and the MAC address as Not available, so no MAC address was reported. The network topology shows **localhost** (the scanning machine) connected to **192.168.56.1**.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

Note: A ping scan only discovers hosts and does not scan ports, so it is quick and low-impact. Host-only and virtual lab networks usually contain very few hosts, so only one live host was found on the whole subnet.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified Apache, WordPress 7.1.4 and WP Download Manager 3.3.76	Attackers may use exposed technology/version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl showed the Apache banner, WordPress caching headers and the redirect plugin in use	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS software version disclosed	DNSRecon showed name servers reporting BIND 9.16.23-RH, plus mail and cPanel service records	Version and service details help build a broader infrastructure profile	● **Medium**
6	Email and DNS hardening gaps	SPF record ends in ~all (soft fail) and WHOIS shows DNSSEC unsigned	Weaker protection against email spoofing and DNS tampering	● **Low**
7	Public contact email exposed	WhatWeb exposed info@networkwalks.com	Can be harvested for phishing or social engineering	● **Low**
8	Live host visible on lab network	Zenmap identified 192.168.56.1 on the lab network	Unknown or unauthorized devices may potentially be present on a network	● **Low**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Reduce publicly exposed version information**
   Remove the WordPress generator tag, minimize web server banners and configure the DNS software so it does not reveal its version.
2. **Keep software updated**
   WordPress core, plugins such as WP Download Manager, and other web technologies should be updated regularly and reviewed against current security advisories.
3. **Review HTTP headers and cookies**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Strengthen DNS and email authentication**
   Review DNS records regularly, enable DNSSEC where supported, and after validating legitimate senders move SPF from ~all to -all and confirm DKIM and DMARC are configured.
5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking program, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

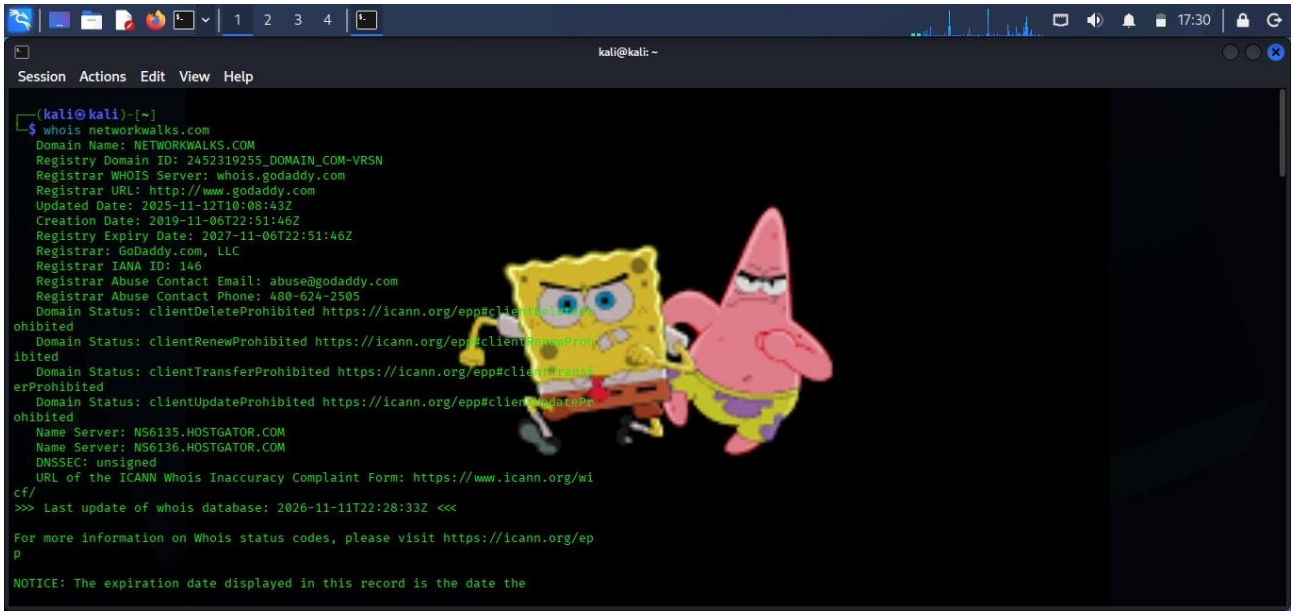
In the network scanning activity, I used Zenmap to discover live hosts on my local lab network, collect IP and MAC address information, and create a network topology.

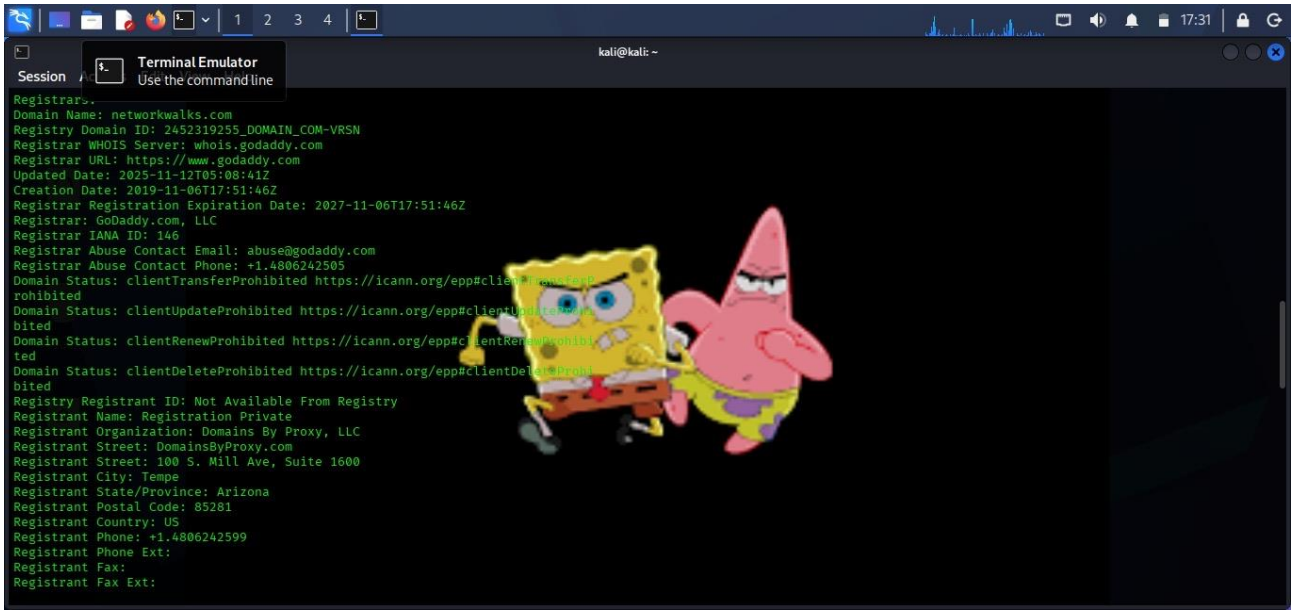
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

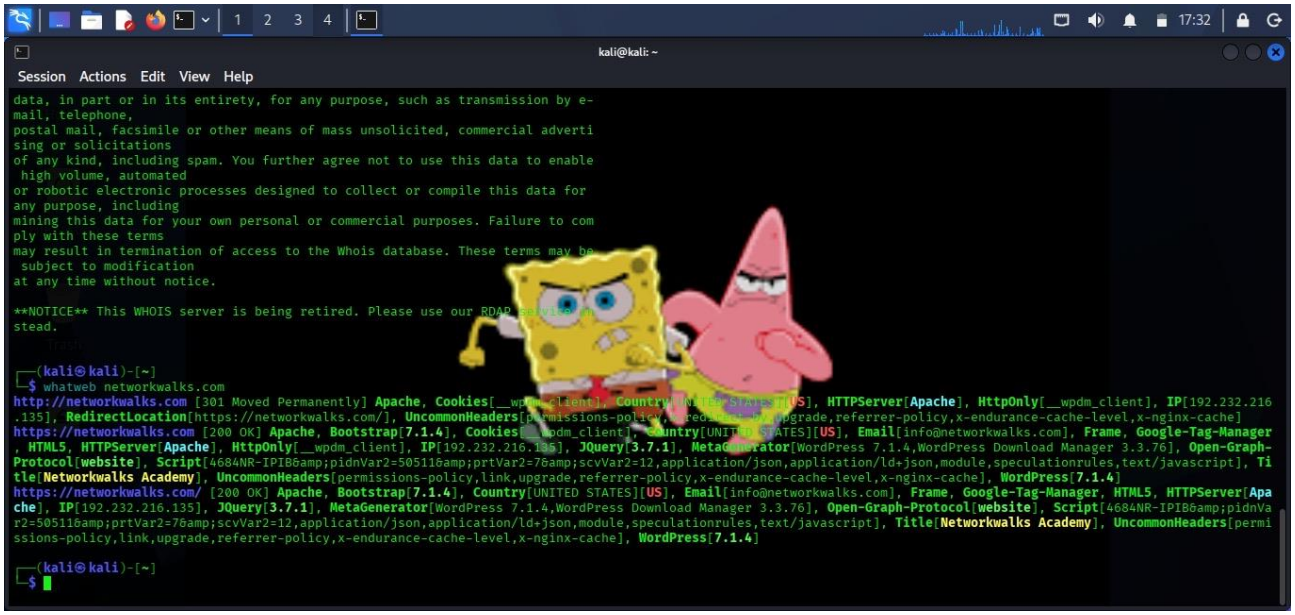
I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

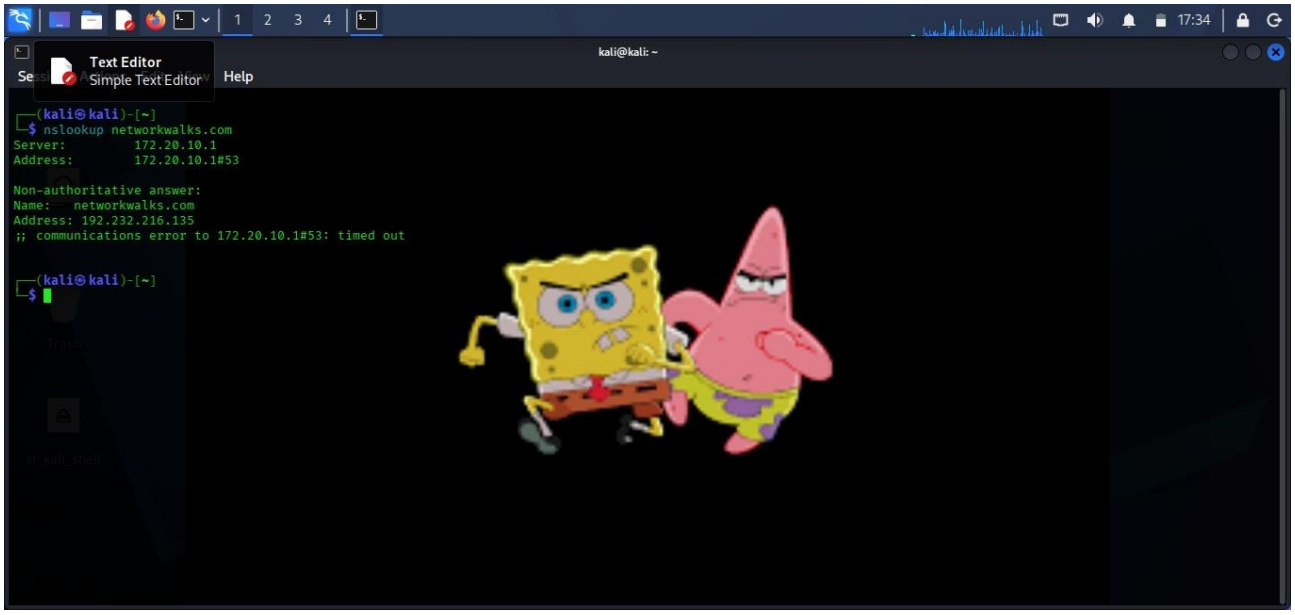
Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

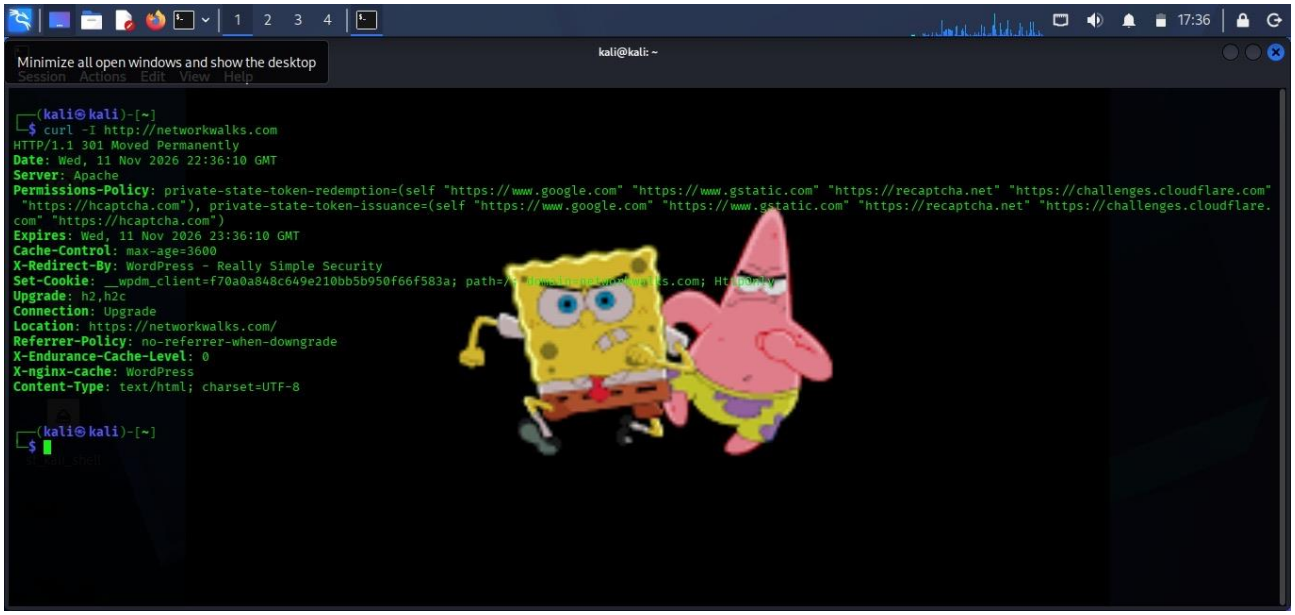
8. Evidences Collected

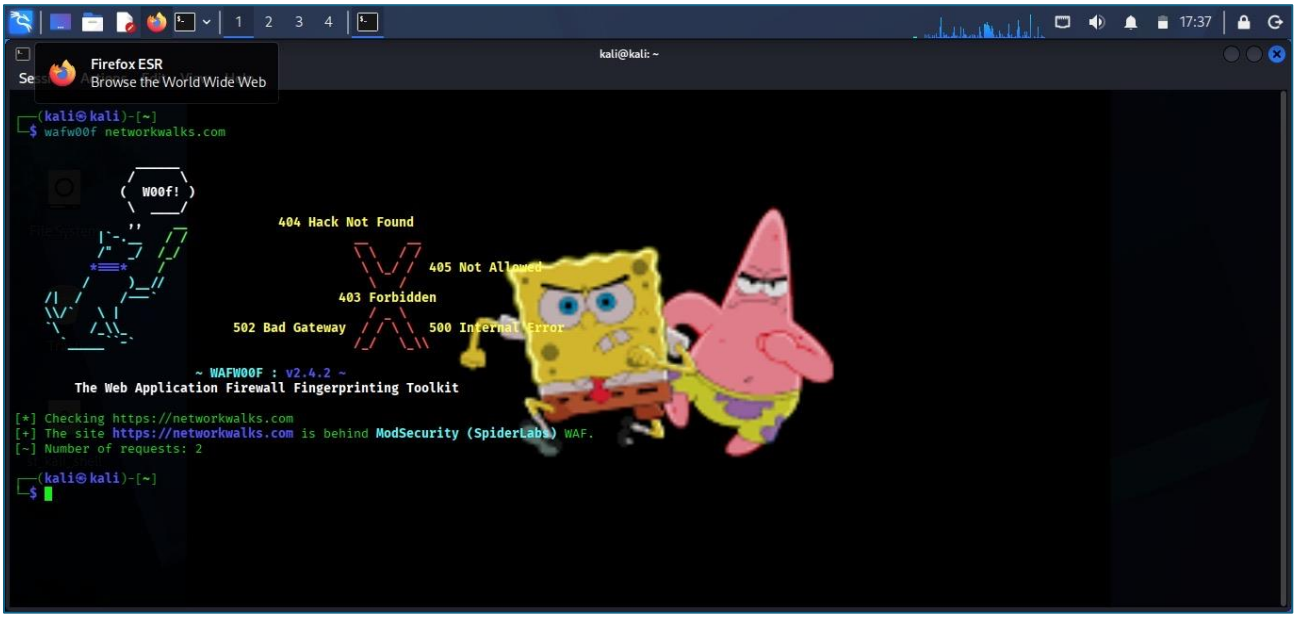


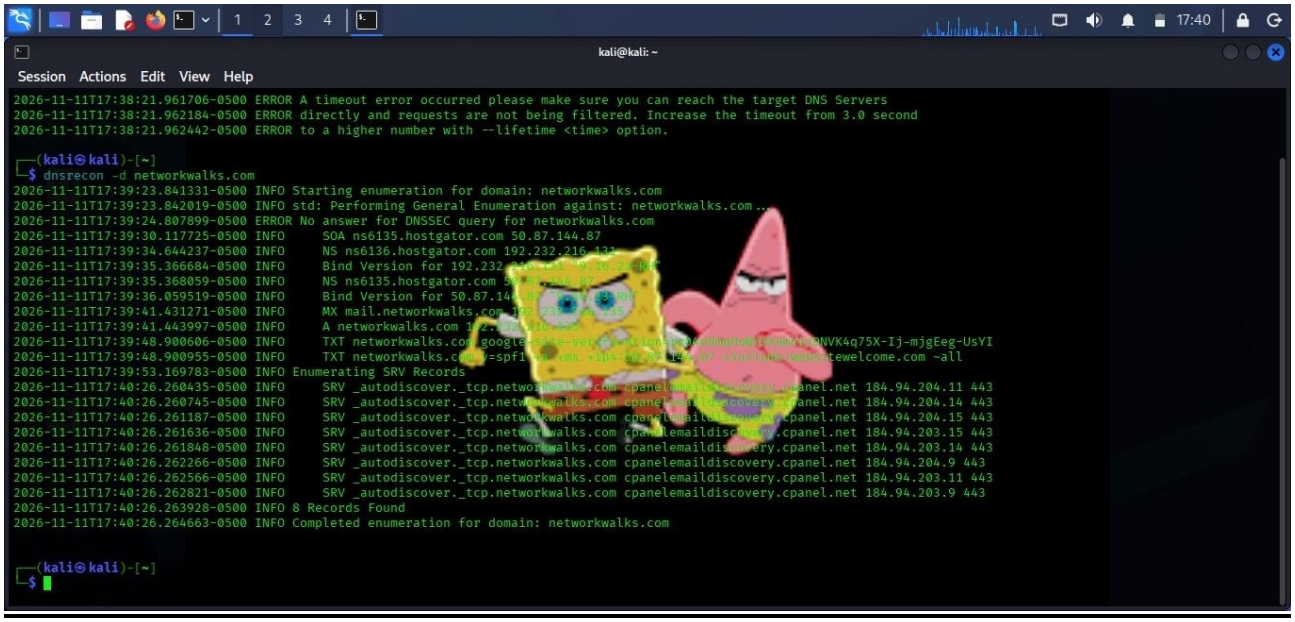


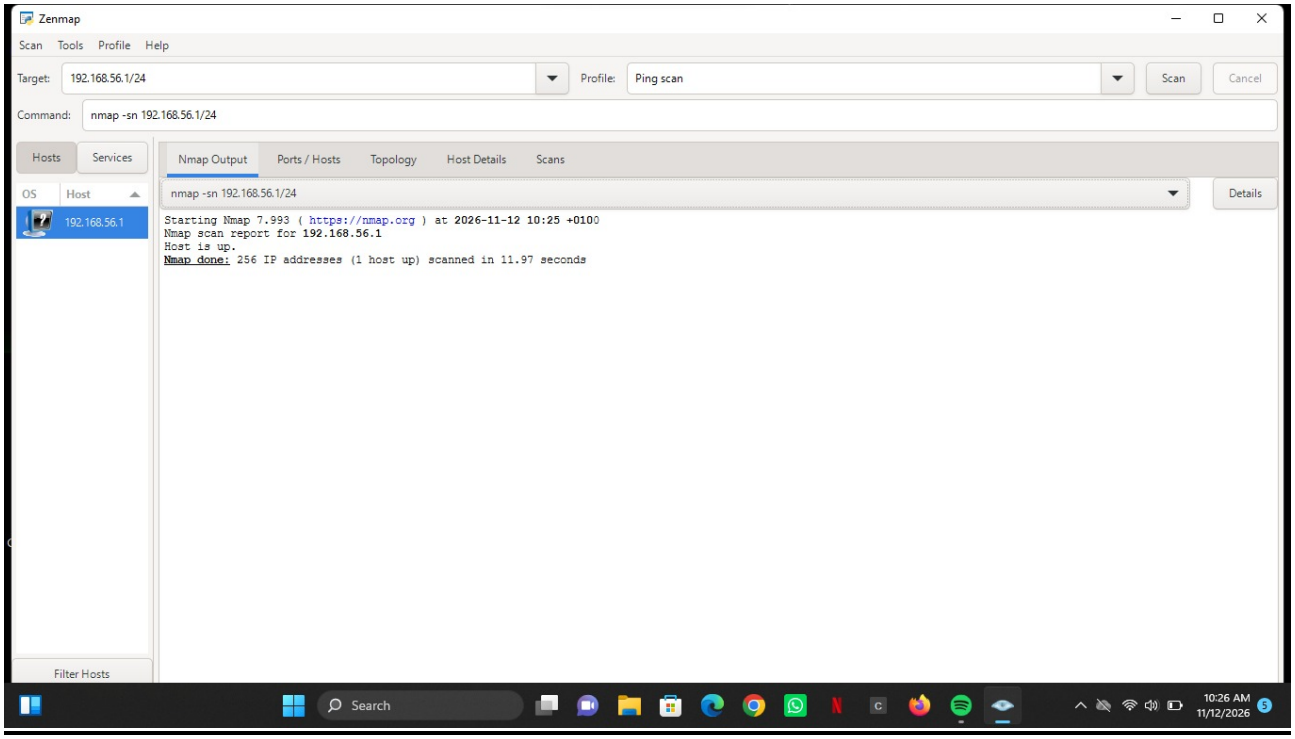

kali@kali: ~
Session Actions Edit View Help
data, in part or in its entirety, for any purpose, such as transmission by e-mail, telephone,
postal mail, facsimile or other means of mass unsolicited, commercial advertising or solicitations
of any kind, including spam. You further agree not to use this data to enable high volume, automated
or robotic electronic processes designed to collect or compile this data for any purpose, including
mining this data for your own personal or commercial purposes. Failure to comply with these terms
may result in termination of access to the Whois database. These terms may be subject to modification
at any time without notice.
NOTICE This WHOIS server is being retired. Please use our RDAP service instead.
(kali㉿kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
(kali㉿kali)-[~]
$

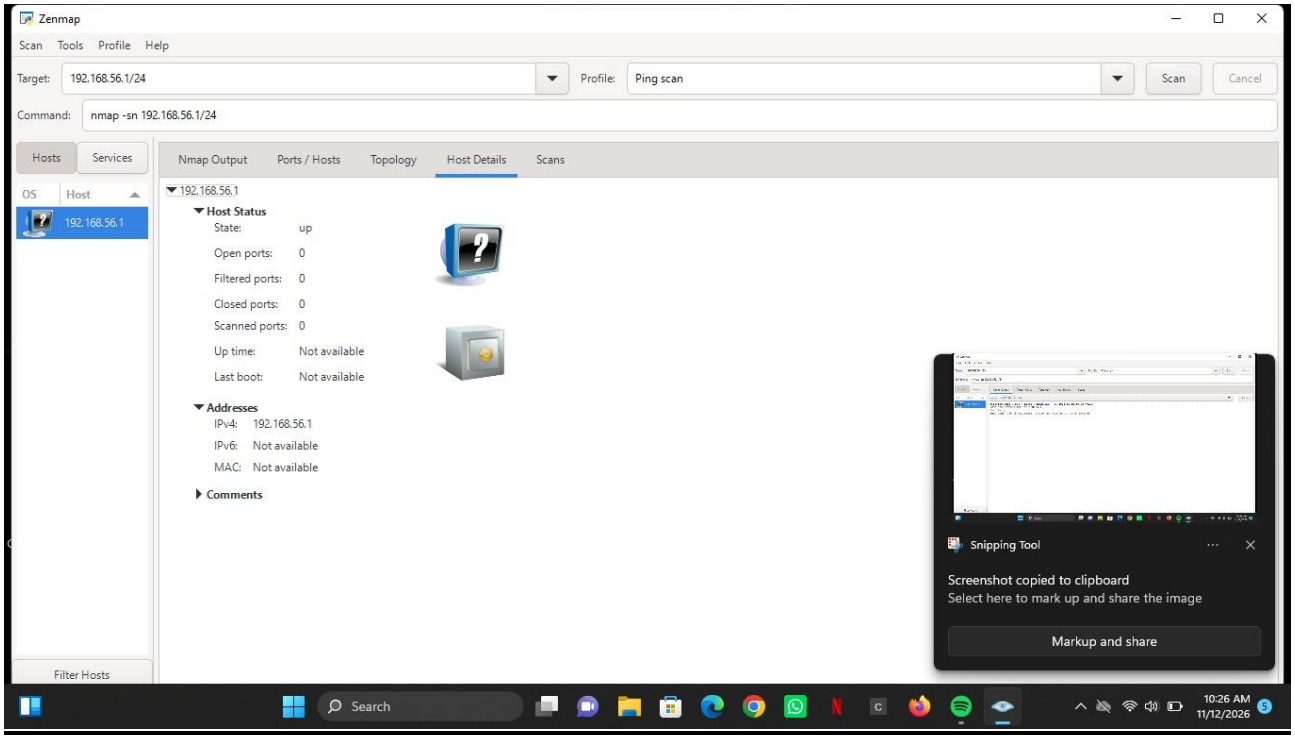

Text Editor
Simple Text Editor
kali@kali: ~
Help
(kali㉿kali)-[~]
$ nslookup networkwalks.com
Server: 172.20.10.1
Address: 172.20.10.1#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
;; communications error to 172.20.10.1#53: timed out
(kali㉿kali)-[~]
$

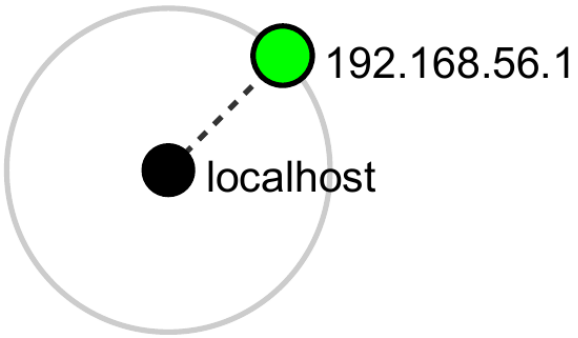

Minimize all open windows and show the desktop
kali@kali: ~
(kali㉿kali)-[~]
$ curl -I http://networkwalks.com
HTTP/1.1 301 Moved Permanently
Date: Wed, 11 Nov 2026 22:36:10 GMT
Server: Apache
Permissions-Policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
Expires: Wed, 11 Nov 2026 23:36:10 GMT
Cache-Control: max-age=3600
X-Redirect-By: WordPress - Really Simple Security
Set-Cookie: __wpdm_client=f70a0a848c649e210bb5b950f66f583a; path=/; Http Only
Upgrade: h2,h2c
Connection: Upgrade
Location: https://networkwalks.com/
Referrer-Policy: no-referrer-when-downgrade
X-Endurance-Cache-Level: 0
X-nginx-cache: WordPress
Content-Type: text/html; charset=UTF-8
(kali㉿kali)-[~]
$


Firefox ESR
Browse the World Wide Web
kali@kali: ~
(kali㉿kali)-[~]
$ wafw00f networkwalks.com
W00f!
404 Hack Not Found
405 Not Allowed
403 Forbidden
502 Bad Gateway
500 Internal Error
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit
[+] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
(kali㉿kali)-[~]
$


kali@kali: ~
Session Actions Edit View Help
2026-11-11T17:38:21.961706-0500 ERROR A timeout error occurred please make sure you can reach the target DNS Servers
2026-11-11T17:38:21.962184-0500 ERROR directly and requests are not being filtered. Increase the timeout from 3.0 second
2026-11-11T17:38:21.962442-0500 ERROR to a higher number with --lifetime <time> option.
(kali㉿kali)-[~]
$ dnsrecon -d networkwalks.com
2026-11-11T17:39:23.841331-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-11T17:39:23.842019-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-11T17:39:24.807899-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-11T17:39:30.117725-0500 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-11-11T17:39:34.644237-0500 INFO NS ns6136.hostgator.com 192.232.216.131
2026-11-11T17:39:35.366684-0500 INFO Bind Version for 192.232.
2026-11-11T17:39:35.368059-0500 INFO NS ns6135.hostgator.com
2026-11-11T17:39:36.059519-0500 INFO Bind Version for 50.87.14
2026-11-11T17:39:41.431271-0500 INFO MX mail.networkwalks.com
2026-11-11T17:39:41.443997-0500 INFO A networkwalks.com
2026-11-11T17:39:48.900606-0500 INFO TXT networkwalks.com
2026-11-11T17:39:48.900955-0500 INFO TXT networkwalks.com
2026-11-11T17:39:53.169783-0500 INFO Enumerating SRV Records
2026-11-11T17:40:26.260435-0500 INFO SRV _autodiscover._tcp.networkwalks.com
2026-11-11T17:40:26.260745-0500 INFO SRV _autodiscover._tcp.networkwalks.com
2026-11-11T17:40:26.261187-0500 INFO SRV _autodiscover._tcp.networkwalks.com
2026-11-11T17:40:26.261636-0500 INFO SRV _autodiscover._tcp.networkwalks.com
2026-11-11T17:40:26.261848-0500 INFO SRV _autodiscover._tcp.networkwalks.com
2026-11-11T17:40:26.262266-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-11T17:40:26.262566-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-11T17:40:26.262821-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-11T17:40:26.263928-0500 INFO 8 Records Found
2026-11-11T17:40:26.264663-0500 INFO Completed enumeration for domain: networkwalks.com
(kali㉿kali)-[~]
$


Zenmap
Scan Tools Profile Help
Target: 192.168.56.1/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 192.168.56.1/24
Hosts
Services
OS Host
192.168.56.1
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn 192.168.56.1/24
Details
Starting Nmap 7.993 (https://nmap.org) at 2026-11-12 10:25 +0100
Nmap scan report for 192.168.56.1
Host is up.
Nmap done: 256 IP addresses (1 host up) scanned in 11.97 seconds
Search
10:26 AM
11/12/2026





-End-

Author
Morakinyo Bolatito
Cybersecurity Professional
LinkedIn: https://www.linkedin.com/in/bolaoluwatito-morakinyo

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Project Information
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** https://github.com/morakinyobolatito-afk/Bolatito-Morakinyo-B084D